

DESIGN & BUILD TUTORIAL

Radio Lab

Build Your Own

AM Station

A tunable AM transmitter and its matching receiver — four transistors and a coil on one board, three transistors and a diode on the other. Every stage is derived from first principles, proven by simulation, and buildable on a perfboard with parts from any hobby shop. You will hear your own voice come out of a radio you tuned yourself.

Designed with tscircuit · React for electronics

29 math checks · 7 simulation proofs · 76 parts · zero DRC errors · October 2026

AM 650-850 KHZ · COLPITTS · ENVELOPE DETECTOR · LM386

Table of Contents

1 · What you are building	3
1.1 · Both boards, at a glance	3
1.2 · How to use this document	4
2 · How radio actually works	5
2.1 · Wavelength, frequency, and why we chose the AM band	5
2.2 · Amplitude modulation — the oldest trick that still works	5
2.3 · Near field vs far field, or why a room and not a city	6
3 · The LC tank — the heart of tuning	7
3.1 · The tuning knob is the lesson	7
3.2 · Q and bandwidth — how sharp the bell is	7
4 · The transmitter, stage by stage	8
4.1 · The microphone and its preamp (MIC1, Q1)	9
4.2 · The modulator (Q2) — a power rail that sings	9
4.3 · The oscillator (Q3) — where the carrier is born	10
4.4 · The modulated final (Q4) — where AM happens	11
4.5 · Reading the schematic	11
5 · Proving it before you build it	12
5.1 · The simulation story, in four pictures	12
5.2 · Two flaws the math caught before you did	14
5.3 · The checks you can rerun anywhere	14
6 · The receiver, stage by stage	15
6.1 · Antenna, tank, and the art of not loading	15
6.2 · Buffer and amplifier (Q1, Q2)	16
6.3 · The detector (D1) — where radio becomes audio	16
6.4 · Volume, the signal LED, and the LM386	17
6.5 · Why not something fancier?	17
7 · Building the pair	18
7.1 · The shopping list	18
7.2 · Perfboard or PCB, both work	19
7.3 · Transistor orientation — the one gotcha	19
7.4 · Wiring the mechanical parts	20

8 · Bring-up: making them talk	20
8.1 · Step 4 deserves a paragraph of its own	21
8.2 · The tuning game	21
9 · Twelve experiments	21
9.1 · The three that teach the most	22
10 · Troubleshooting	23
10.1 · The two-minute sanity loop	23
11 · The rules, the safety, and the road ahead	24
11.1 · Transmitting legally	24
11.2 · Safety, briefly	24
11.3 · Where this road goes next	24

1 • What you are building

Radio Lab is a pair of small circuit boards: an **AM transmitter** that turns your voice into a radio signal, and an **AM receiver** that plucks that signal back out of the air. The transmitter has a microphone, four transistors, a coil, and a tuning knob. The receiver has a wire antenna, three transistors, a germanium diode, an audio amplifier chip, a volume knob, and its own tuning knob. Both run for days on ordinary 9 V batteries. Everything is through-hole, everything sits on a 0.1 inch grid, and every part is the kind you can buy from any hobby electronics shop for pocket change.

The point of the pair is the ritual at the heart of real radio. You speak into the transmitter, turn its TUNE knob to pick a quiet spot on the medium-wave band, and then walk the receiver's knob slowly across the dial until — through a wall of static — your own voice suddenly appears in the speaker. That moment, when a signal you created with your own hands crosses the room as an electromagnetic wave, is what this project is really about. Everything in this document exists to make that moment inevitable rather than lucky.

AM-TX transmitter · 36 parts · 91×56 mm	AM-RX receiver · 40 parts · 99×61 mm	633-836 kHz the transmitter's tuning range
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1.1 • Both boards, at a glance

The two renders below are generated straight from the design source. The transmitter reads left to right like its schematic: power and microphone at the left, audio preamp and modulator in the middle, the oscillator and its tank coil on the right, and the antenna wire pad at the far right edge. The receiver follows the same discipline: antenna and tank at the left, the two radio-frequency transistor stages in the middle, and the detector, volume control, and LM386 audio amplifier filling the right half. Every placement sits on the 2.54 mm perfboard grid, so if you do not want to order circuit boards, you can copy these layouts hole-for-hole on perfboard.

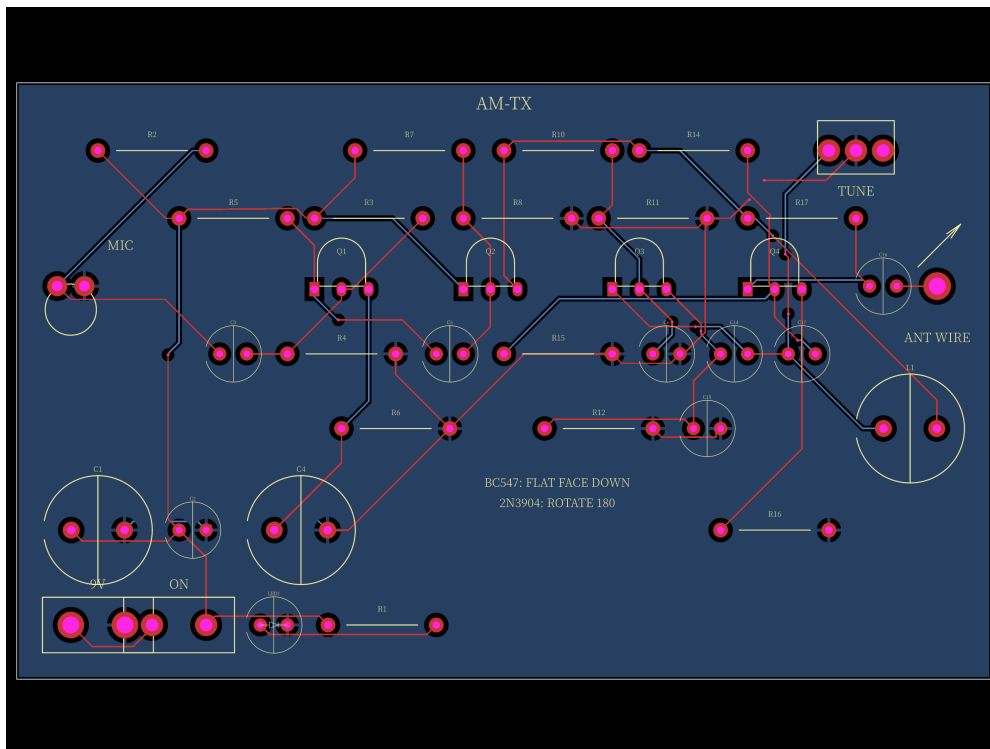


Figure 1-1 — The AM-TX transmitter board (tscircuit PCB render). Bottom-layer ground pour not shown.

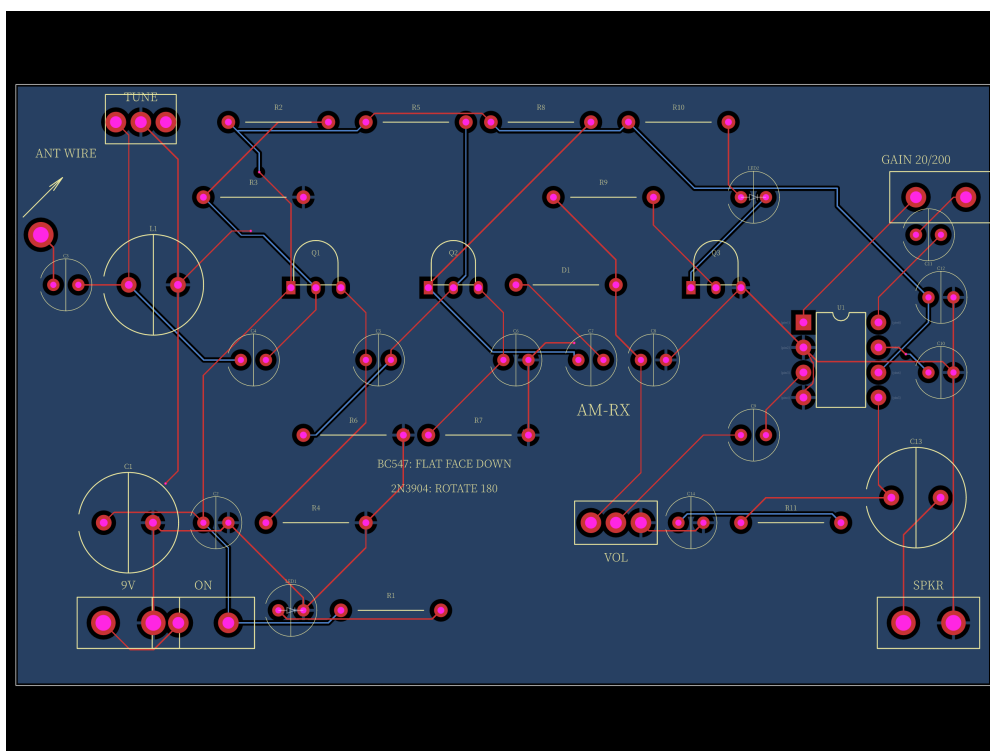


Figure 1-2 — The AM-RX receiver board. Note the big tuning capacitors (CV1) and the volume potentiometer pads, both meant to be wired to panel-mount parts.

1.2 • How to use this document

Chapters 2 and 3 teach the three ideas the whole project hangs on: what an AM signal is, and why a coil-plus-capacitor *tank circuit* is the device that both creates and selects radio signals. Chapters 4 and 6 walk

every stage of the transmitter and receiver with the real component values and the arithmetic that chose them. Chapter 5 is the part most tutorials skip: we prove the oscillator works by numerically integrating its differential equations, and we show you the two design flaws that the simulation caught before they could become a dead board on your desk. Chapters 7 through 11 are the practical half: the shopping list, the build, the bring-up ritual, twelve experiments, a fault-finding guide, and the legal picture.

You do not need any radio background. You should be comfortable identifying resistor color codes, and ideally you have soldered a handful of kits before. Wherever a step has a common failure mode, it is called out in a box like the design notes you will meet along the way.

2 • How radio actually works

A radio wave is what happens when a changing electric current launches a ripple of electric and magnetic fields that then travels on its own, at the speed of light, without needing any wire to carry it. Accelerate electrons in a wire — that is, drive an alternating current into it — and a small fraction of the energy you pour in never comes back down the wire: it leaves as radiation. A receiving antenna is just the same process in reverse. The passing wave pushes electrons in it back and forth, and that tiny induced current is the raw material every receiver ever built has to work with.

2.1 • Wavelength, frequency, and why we chose the AM band

Frequency and wavelength are two views of the same thing: multiply them and you get the speed of light. At the frequencies this project uses — around 700 kilohertz — one wave cycle stretches about 400 meters from crest to crest. That single fact shapes everything practical about the design. A transmitting antenna works best when it is a quarter of the wavelength long; at 400 m that would be a 100 m tower. Our 1-2 metre wire is, by radio standards, almost nothing — an *electrically short* antenna that behaves mostly like a small capacitor. It radiates only a whisper of the transmitter's power, which is exactly what we want: enough to cross a room, little enough to stay legal and polite.

The medium-wave AM band (roughly 530 to 1700 kHz worldwide) is the perfect classroom. The wavelengths are long enough that circuits are forgiving — a hand-wound coil or a cheap choke works fine, and component tolerances barely matter — yet the band is alive with real broadcast stations you can hunt with the receiver. In Vietnam the national and provincial AM services sit between roughly 540 and 750 kHz (VOV1 on 630 and 675 kHz in most provinces, VOV2 nearby), which is why the transmitter's tuning range, 633-836 kHz, deliberately brackets them: you will be able to hear the real thing and your own signal on the same dial.

2.2 • Amplitude modulation — the oldest trick that still works

A pure carrier wave carries no information; it is just a monotone hum. To send a voice, we let the voice control the carrier's *amplitude* — the size of its swings — while leaving its frequency alone. That is amplitude modulation, AM, the method of the first broadcast era and still the easiest to understand, transmit,

and receive. The transmitter multiplies a fast wave by a slow one; the receiver, incredibly, only needs a diode and a capacitor to undo it, as chapter 6 will show.

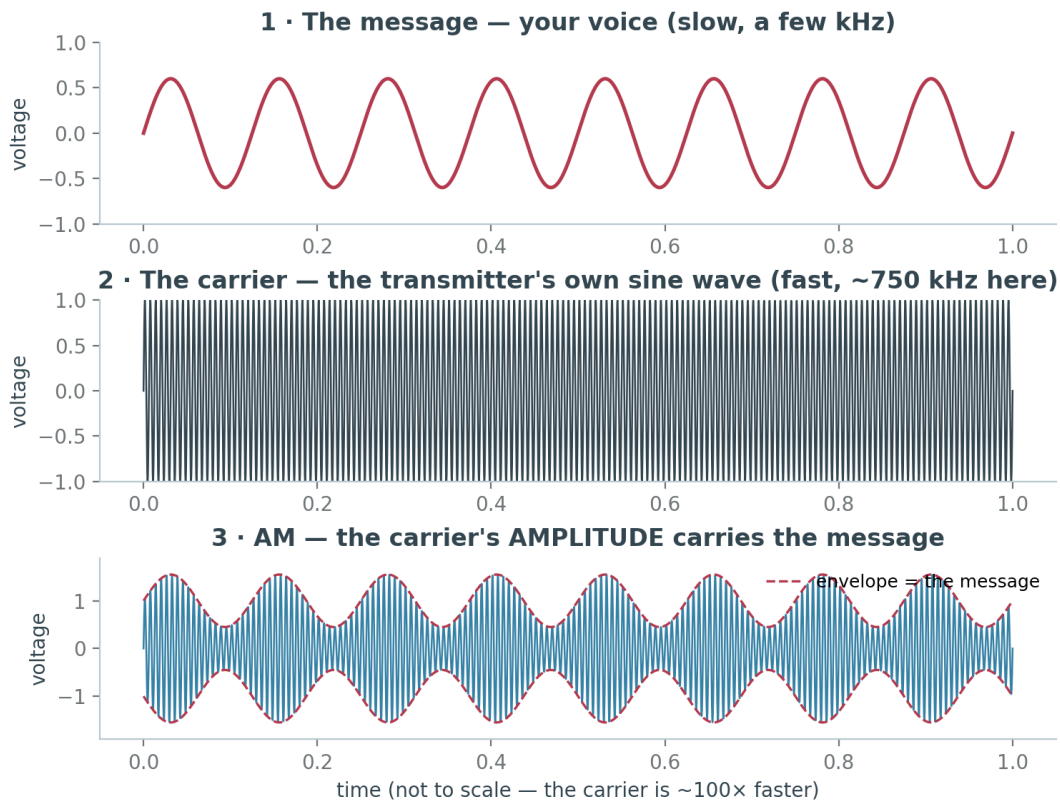


Figure 2-1 · Amplitude modulation. The message (top) rides on the carrier's envelope (bottom). The dashed curves trace that envelope — the receiver's whole job is to recover them.

The depth of modulation, m , says how hard the voice pushes: $m = 0$ is a dead-level carrier, $m = 1$ means the envelope briefly touches zero on voice peaks, and pushing past 1 is *overmodulation* — the carrier momentarily vanishes, and the recovered audio crackles nastily. Our transmitter reaches $m \approx 0.2\text{--}0.35$ in simulation with normal speech, and can be pushed into deliberate overmodulation by shouting — which is one of the experiments in chapter 9, not a defect.

$$m \approx 0.23$$

modulation depth measured on the simulated final stage — comfortable listening range, shouting pushes it toward 1 for the overmodulation demo

2.3 · Near field vs far field, or why a room and not a city

Within about a tenth of a wavelength — here, the first 30–40 metres — the energy around a short antenna has not fully detached into a travelling wave yet. In this *near field*, coupling between two short wires falls off roughly with the square of distance, which is brutally fast. That is the physics behind the project's honest expectation: across a room, the receiver hears you loudly; a few rooms away, faintly; across town, never. Real broadcast stations solve this with kilowatts and hundred-metre towers. You get a house-sized bubble of radio,

which is precisely the right size for learning in — and, not coincidentally, the size that unlicensed low-power rules around the world are written for (chapter 11).

3 • The LC tank — the heart of tuning

Take a coil of wire (an inductor, L) and a capacitor (C), connect them in parallel, and you have built the most important circuit in radio. Charge the capacitor and it dumps its energy into the coil, building a magnetic field; the field then collapses and pushes the energy back into the capacitor. Left alone, this seesaw rings at one precise frequency, $f = 1/(2\pi\sqrt{LC})$, the way a bell rings at one note. Feed the tank a mixture of many frequencies and it answers with a large voltage only at its own note — every other frequency is conducted harmlessly to ground. One circuit, two careers: in the transmitter it *creates* the carrier; in the receiver it *selects* the station.

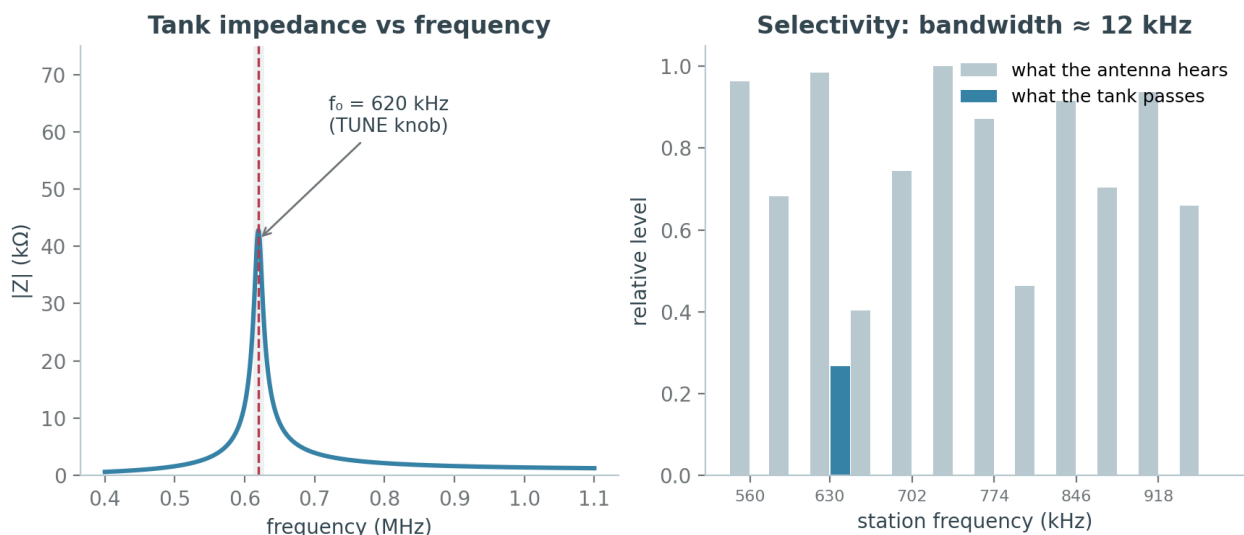


Figure 3-1 · Left: the tank's impedance peaks at its resonant frequency — turning the TUNE capacitor slides that peak across the band. Right: what that means in practice — of all the stations the antenna hears, only the one under the peak survives.

3.1 • The tuning knob is the lesson

The tuning capacitor CV1 in both boards is a *polyvaricon* — the little plastic-dielectric variable capacitor found in every pocket AM radio, two metal combs meshing as you turn the shaft, 10 to 280 pF. Because f depends on the square root of C , that 28:1 capacitance swing becomes roughly a 1.5:1 frequency swing on the transmitter (633 to 836 kHz) and wider on the receiver. Turn the knob and you literally watch the station you have selected rise and fall in the noise — resonance as a physical, audible experience. When someone asks you what the tuning knob on a radio does, after this project you will be able to say: it slides a bell's note across the whole band until the note matches the station.

3.2 • Q and bandwidth — how sharp the bell is

The quality factor Q measures how many times the energy circulates before being lost, and it sets the width of the peak: $\text{bandwidth} = f/Q$. Our receiver's tank idles near $Q \approx 50$, giving about 15 kHz of bandwidth around a 750 kHz carrier — comfortably wide enough to carry the 5 kHz of audio an AM station uses, and narrow enough to reject the neighbouring channel 9 kHz away. Everything the design does around the tank — the light 47 pF antenna coupling, the 1 M Ω /470 k Ω bias pair, the buffer transistor that follows — is in service of not loading it, because every bit of load widens the bell and lets neighbours bleed in. You can hear this directly: experiment 9 in chapter 9 has you load the tank on purpose and listen to the selectivity collapse.

Quantity	Transmitter tank	Receiver tank
Coil L1	100 μ H radial choke	220 μ H radial choke
Capacitance	340 pF fixed + CV1 (10-280 pF)	CV1 (10-280 pF) + strays
Frequency range	633 - 836 kHz (ideal)	$\approx$ 614 kHz - 1.8 MHz
Role	creates the carrier	selects the station
Loaded by	feedback pair + final stage	antenna tap + buffer (kept light)

Table 3-1 · The same idea, used twice. The transmitter's 340 pF fixed component is the series pair of the two 680 pF feedback capacitors — see chapter 4.

$$f = 1/(2\pi\sqrt{LC})$$

the one formula this whole project turns on —
the bell's note

4 · The transmitter, stage by stage

The transmitter is four transistors arranged like a miniature broadcast station. Q1 makes the microphone's whisper loud. Q2 turns that audio into a power supply that wobbles with your voice. Q3 rings a tank circuit to make a clean, steady carrier — the station's frequency, set by the TUNE knob. Q4 is the final amplifier, driven hard enough that its output can only slam between ground and the wobbly supply, which stamps the voice onto the carrier as amplitude modulation. This chapter walks each stage with its real values and the arithmetic behind them.

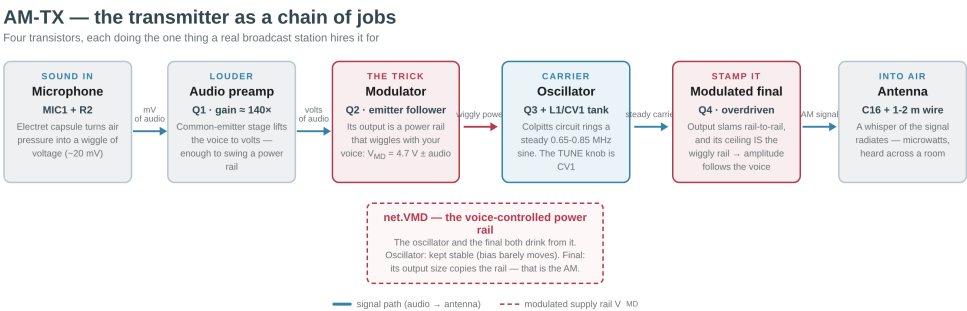


Figure 4-1 · The transmitter as a chain of jobs. Blue is the signal path; the dashed red rail is the modulated supply — the trick that does the amplitude modulating.

4.1 · The microphone and its preamp (MIC1, Q1)

An electret capsule is a tiny metal can with a permanently charged film and a built-in field-effect transistor that needs power. R2 (2.2 kΩ) supplies that power from the 9 V rail; the capsule's output pin then wiggles a few volts below the rail carrying your voice as perhaps 20 mV of AC. C3 passes the AC to Q1's base and blocks the DC. Q1 is a textbook common-emitter amplifier: R3 and R4 hold its base at 1.15 V, R6 sets its current to about 0.8 mA, and R5 is its load. Because C4 short-circuits R6 at audio frequencies, the stage's gain becomes $R5/r_e \approx 4.7\text{ k}\Omega/33\text{ }\Omega \approx 140\times$. The 20 mV voice becomes volts — enough to swing a power rail, which is exactly what the next stage needs.

Part	Value	Why
R2	2.2 kΩ	powers the electret's internal FET (~0.3 mA)
C3	1 μF film	passes voice, blocks DC; film = no polarity
R3/R4	68 k / 10 k	base at 1.15 V — mid-current bias point
R5	4.7 kΩ	collector load; sets gain together with r _e
R6 + C4	560 Ω + 100 μF	R6 sets 0.8 mA; C4 makes it invisible at audio → gain 140×

Table 4-1 · The audio front end. V_C idles at 5.2 V — nearly mid-rail, with clean headroom in both directions for voice peaks.

4.2 · The modulator (Q2) — a power rail that sings

Q2 is an emitter follower with one unusual habit: nobody uses its output as a *signal*. Its base is biased at 5.4 V by R7/R8 and receives the audio from Q1 through C5; its emitter therefore sits a diode-drop below and *follows* the audio. Everything downstream drinks from that emitter — net.VMD, a supply rail that idles at

4.7 V and swings several volts with your voice. Emitter followers can supply current easily (the oscillator and final together draw about 1.6 mA), so the rail stays stiff at radio frequencies while it sways at audio ones. Hold this picture: **the modulator is the power supply, and the power supply is the modulator.**

$$V_{MD} = 4.7 \text{ V} \pm \text{audio}$$

idle level of the modulated rail — swings 3.2 to 6.2 V on speech peaks (deeper if you shout)

4.3 · The oscillator (Q3) — where the carrier is born

Q3 is a common-base Colpitts oscillator, the circuit that has started more radio signals than any other. Its collector tank — L1 (100 μ H) in parallel with CV1 and the series pair C14/C15 — rings at the frequency from chapter 3. The feedback path is the capacitive divider C14/C15 between collector and emitter: the emitter sits at their midpoint, so a measured slice of the collector's swing feeds back in phase. The amplifier then returns more energy each cycle than the tank loses, and the oscillation grows until the transistor's own base-emitter junction begins rectifying on peaks — a gentle, self-adjusting brake that pins the carrier at a steady ~ 1.3 V amplitude. Designers call this self-limiting; it is why the carrier is stable without any regulation.

Two details deserve their own paragraphs, because both are places where plausible designs quietly die. First, the coil: L1 hangs between the collector and the VMD rail, not ground. A coil is a short circuit at DC, so it doubles as the collector's power feed while behaving as a high impedance (about 470 Ω at 750 kHz) to the radio signal — feed and resonator in one part. Our first draft had the coil to ground with a resistor feeding the collector, which pins the collector at 0 V and kills the stage instantly; the simulation caught it (chapter 5). Second, the base bypass C6 is deliberately small (2.2 nF). It must be a short circuit at radio frequencies so the base sits still — but if it were the habitual 100 nF, it would also short-circuit the audio-rate component of the bias, and the stage's operating point would wander with the voice. 2.2 nF is 45 Ω at 750 kHz and 36 k Ω at 2 kHz: stiff where it must be, transparent where it must be.

Part	Value	Why
L1	100 μ H	tank coil AND collector feed in one part
CV1	10-280 pF	the TUNE knob: 633-836 kHz
C14/C15	680 pF $\times 2$	feedback divider; series pair adds 340 pF to the tank
R10/R11	27 k / 10 k	base bias at 1.27 V, hung from the wobbly rail — on purpose
C6	2.2 nF	RF ground for the base, audio-transparent (see text)
R12	1 k Ω	emitter current ≈ 0.6 mA $\rightarrow$ gm ≈ 22 mS

Table 4-2 · The oscillator. Why 680 pF and not the textbook 220 pF pair? Chapter 5 — the weak-feedback failure is the project's best story.

4.4 · The modulated final (Q4) — where AM happens

Broadcast engineers call it high-level modulation: the final amplifier's output can never exceed its supply rail, so if you make the *rail* dance to the music, the *output* dances with it. Q4 is driven from the tank through C17, biased by R14/R15, and degenerated by R16 — giving it a small-signal gain of $3.8\times$, deliberately more than enough that the 1.3 V carrier drives it into full swing. Its collector therefore slams between about 0.3 V and the VMD rail on every carrier cycle: a squared-up carrier whose height is the rail voltage. Voice up, rail up, carrier taller; voice down, rail down, carrier shorter. That is amplitude modulation, stamped on at the last possible instant, exactly as a 50 kW transmitter does it — with a 2N3904 and a 9 V battery instead.

C16 then leaks a measured sliver of this signal into the antenna wire: at 750 kHz its 100 pF is about 2 k Ω of impedance, light enough that the antenna (itself barely 30 pF of capacitance to the world) never loads the final enough to matter. The radiated power is microwatts. If you want more range, the honest route is a better antenna, not more power — and the honest antenna for 400 m wavelengths is a longer wire, which is also the direction the law frowns on (chapter 11). Enjoy the room-sized bubble; it is the size learning comes in.

4.5 · Reading the schematic

The full schematic below is rendered directly from the board source, and the source file is written to be read top to bottom the same way: every wire in the design exists as an explicit trace in `transmitter/index.tsx`, grouped under banner comments that mirror these sections. If you ever wonder what something connects to, the answer is one `grep` away — the netlist and the documentation cannot drift apart, because they are the same text.

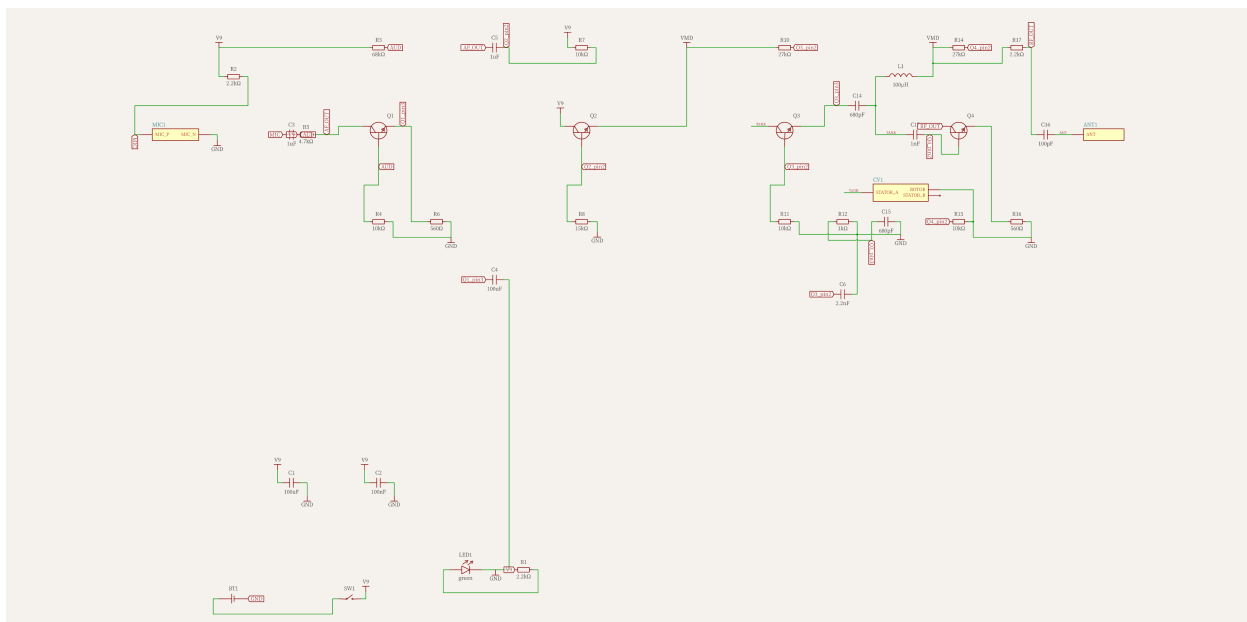


Figure 4-2 · The AM-TX schematic (tscircuit render). Signal flows left to right; net labels V9, VMD, TANK and GND mark the shared rails.

5 • Proving it before you build it

Most electronics tutorials ask you to trust them. This chapter asks you to trust arithmetic. The repository ships two programs that check the design the way an engineer would: `calc/circuit_math.mjs` re-derives every bias point, gain, frequency and time constant from the component values and asserts 29 facts about them; `calc/sim_colpitts.mjs` goes further and numerically integrates the oscillator's differential equations — a real transistor model, the real bias network, Runge-Kutta at 2-nanosecond steps — and watches it wake up, tune, and modulate. Both exit non-zero if any assertion fails. Run them yourself: `npm run math` and `npm run sim`.

5.1 • The simulation story, in four pictures

The first test seeds the tank with 1 millivolt — thermal noise scale — and lets time run. In a healthy oscillator, loop gain above unity amplifies that seed exponentially until the self-limiting brake engages; that is precisely what figure 5-1 shows: from nothing to a full ~1.3 V carrier in about two hundred microseconds. The second test measures the frequency by counting zero crossings and compares it with the textbook formula: within 2% across the whole band (figure 5-2). Turning CV1 through its range slides the carrier monotonically from 847 to 666 kHz — the knob works, provably.

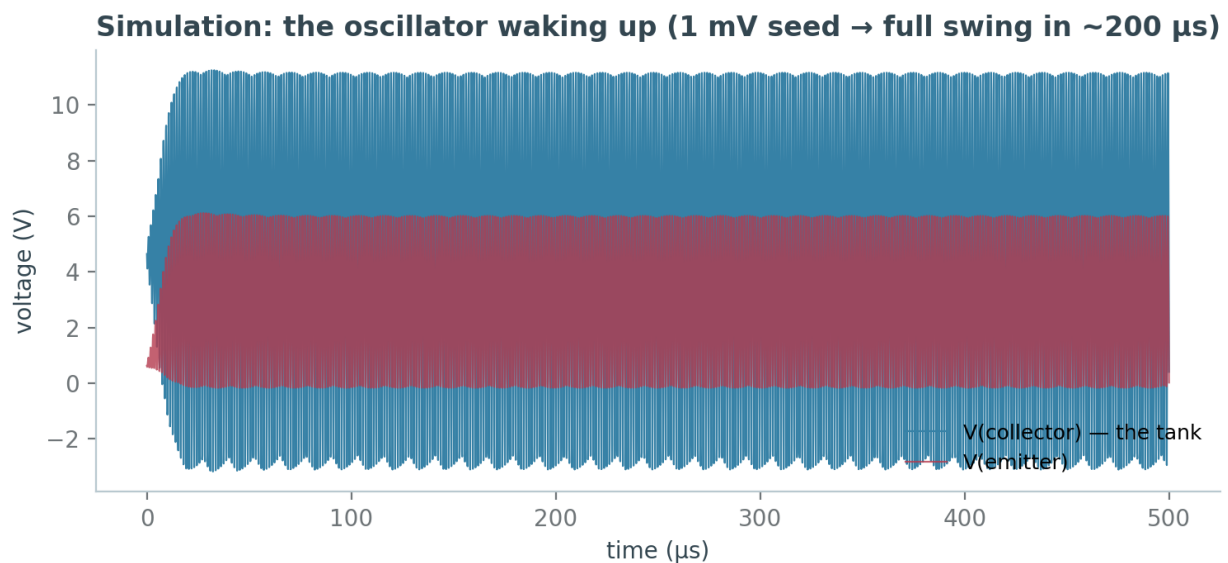


Figure 5-1 • Simulated start-up. A 1 mV seed grows into a full carrier — the definition of an oscillator that starts.

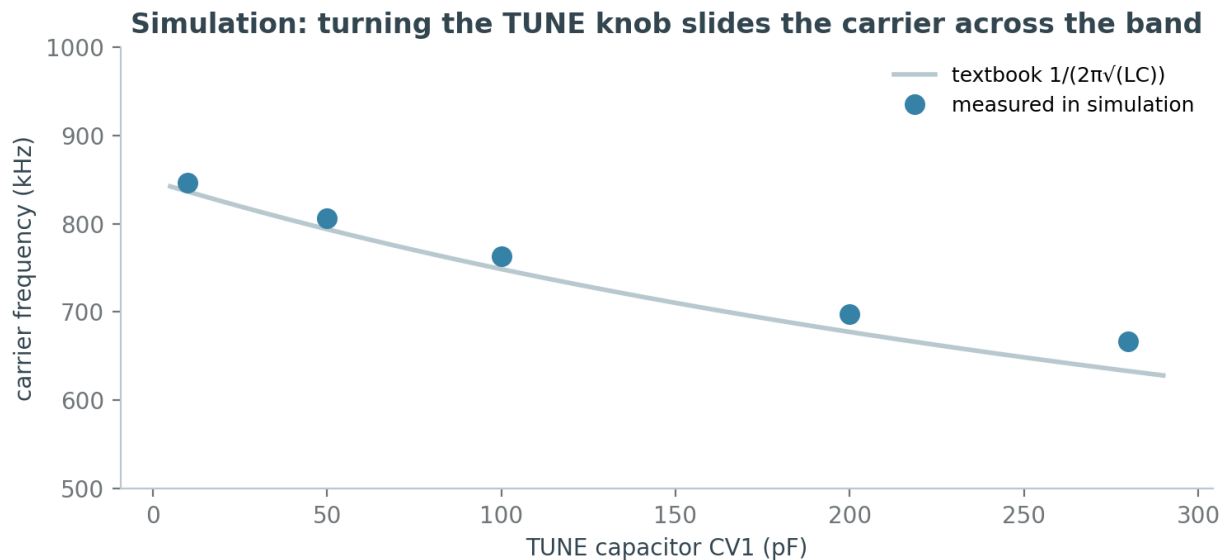


Figure 5-2 · The TUNE sweep, measured in simulation vs the ideal formula. The small offset is the emitter's loading pulling the tank slightly — real physics, honestly reported.

The remaining two pictures are the AM proof. Figure 5-3 shows the oscillator itself while its supply rail sings a 2 kHz sine: the carrier never drops out and its envelope wobbles only mildly — a clean, steady carrier, which is what you want from an oscillator. Figure 5-4 shows the final amplifier's output under the same conditions: the envelope now visibly breathes with the rail, tracking it closely. The measured modulation depth, $m \approx 0.23$, matches the design arithmetic. The voice is on the air.

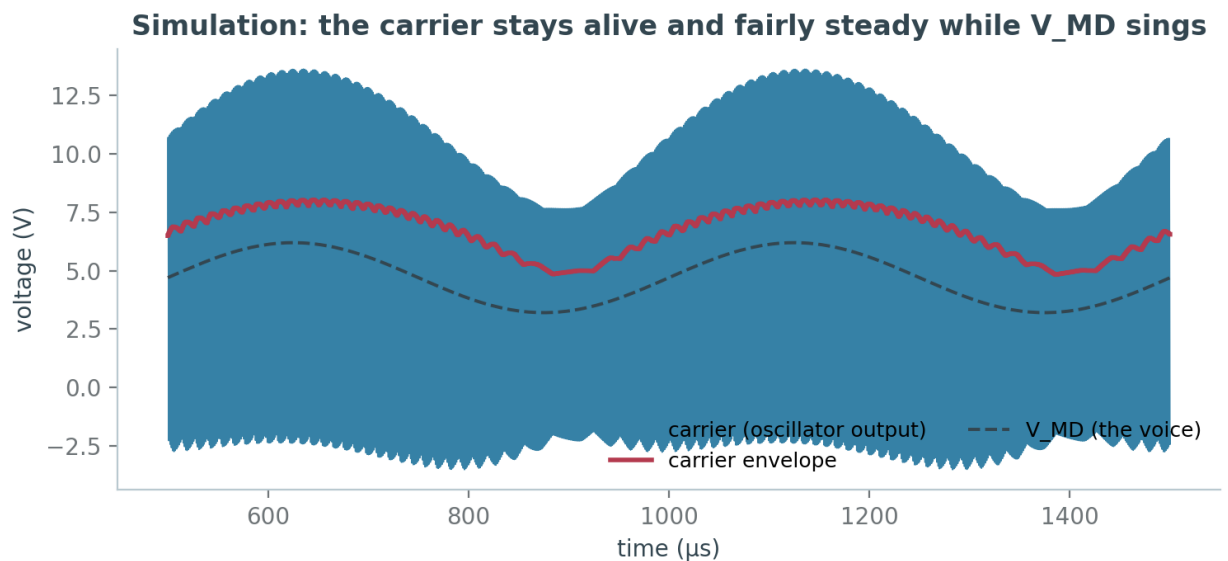


Figure 5-3 · The oscillator under a singing rail: alive, steady, mildly perturbed. Exactly the temperament a carrier should have.

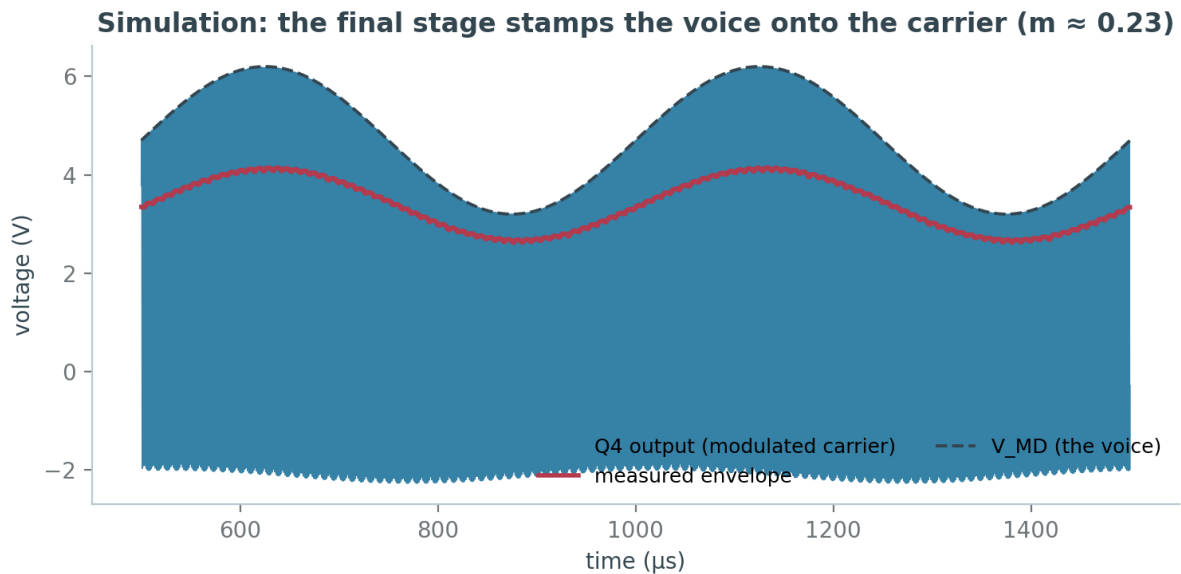


Figure 5-4 · The final stage's output: the envelope (red) chases V_MD (dashed) — amplitude modulation, demonstrated rather than promised.

5.2 · Two flaws the math caught before you did

Both caught flaws are instructive enough to keep on display. The first was the coil-to-ground topology described in chapter 4 — a wiring subtlety that renders the oscillator permanently saturated. No amount of bench debugging would have made it work; it was wrong on paper, and the simulation said so in milliseconds. Finding it cost one evening; finding it with a soldering iron would have cost the whole project.

The second was the feedback divider. The textbook Colpitts drawing shows a symmetric capacitor pair, and 220 pF/220 pF looks perfectly reasonable at 1 MHz. But the start-up condition is not "some feedback" — it is that the transistor-and-divider network must present a negative resistance smaller in magnitude than the tank's losses. At 0.6 mA the 220 pF pair presents about $-20\text{ k}\Omega$ against a $\sim 20\text{ k}\Omega$ tank: ratio 1.0, and in practice, nothing. The 680 pF pair presents about $-2.3\text{ k}\Omega$ against 5.2 k Ω of loaded tank loss — a 2.2 $\times$ margin at mid-band, 1.6 $\times$ even at the hardest corner. That is the difference between an oscillator and a decoration, and it is a difference you can now *compute*, not guess.

29 / 29 design-math assertions passing	7 / 7 simulation tests passing	0 DRC errors on either board
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5.3 · The checks you can rerun anywhere

Everything above is reproducible from the repository with three commands and no test equipment. This matters beyond diligence: it makes the tutorial falsifiable. If you change a component value — and chapter 9 will make you want to — rerun the math and the simulation and watch which assertions move from pass to fail. That feedback loop, values $\rightarrow$ equations $\rightarrow$ verdicts, is the actual skill this project teaches; the radio is just the excuse.

the full proof suite, from a fresh clone

```

$ cd circuit/radio-lab
$ npm run math      # 29 design-equation checks
RESULT: 29 passed, 0 failed
$ npm run sim       # RK4 integration: start-up, tuning, AM
RESULT: 7 passed, 0 failed
$ npm run verify    # board audit: parts, nets, traces, DRC
RESULT: ALL CHECKS PASSED (both boards, zero errors)

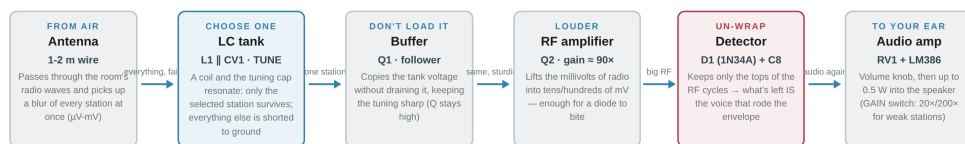
```

6 • The receiver, stage by stage

A receiver's job is a chain of rescues. The antenna hears every station at once, each buried in static; the tank rescues one of them; the buffer keeps the rescue intact; the amplifier makes it big enough to bite a diode; the diode unwraps the voice; the LM386 makes it loud. This design is a TRF — tuned radio frequency — receiver, the direct ancestor of every radio you have owned, stripped to parts you can hold in your head all at once.

AM-RX — the receiver as a chain of rescues

Each stage saves the signal from one problem until audio comes out of the speaker



Side branch: the detector's DC level also feeds Q3 + LED2 — the "signal LED" that glows when you are tuned into a strong station, which turns finding the transmitter into a visual game.

Figure 6-1 • The receiver as a chain of rescues. Each stage exists because of one specific problem; when you troubleshoot, you test the rescues in this order.

6.1 • Antenna, tank, and the art of not loading

The antenna is 1-2 metres of any insulated wire soldered to the ANT pad. At these wavelengths it is essentially a 30 pF capacitor hanging in the air, and the antenna coupling capacitor C3 (47 pF) deliberately connects it to the tank *loosely*: the tank keeps its high Q, and only a fraction of the picked-up signal is traded away. The tank itself is L1 (220 µH) in parallel with CV1 — the TUNE knob — covering roughly 614 kHz to 1.8 MHz: the whole medium-wave band plus margin, completely covering the transmitter's range with room to spare. What the tank does not select, it conducts to ground; the selected station's voltage appears across the coil.

C4 then hands that voltage to Q1 while blocking the coil's DC path (a coil is a short at DC — without C4 the bias network below would be shorted to ground). The bias pair R2/R3 is enormous by amplifier standards — 1 M Ω and 470 k Ω — so that the tank keeps ringing as if almost nobody were listening. This is the chapter-3 lesson in component form: selectivity is Q, and Q is *freedom from load*.

6.2 • Buffer and amplifier (Q1, Q2)

Q1 is an emitter follower: voltage gain slightly below one, but input impedance in the hundreds of kilohms and output impedance around 400 Ω . It copies the tank's voltage without draining it — the polite listener that lets the bell keep ringing. Q2 then does the heavy lifting: a common-emitter stage biased at about 1 mA by R5/R6/R7, with C6 making its emitter an AC ground so its gain is $R8/r_e \approx 2.2\text{ k}\Omega/26\text{ }\Omega \approx 90\times$. A station arriving as 1 mV of RF leaves as 90 mV — enough that the next stage, a plain diode, can finally act like a switch instead of a soft shoulder.

Part	Value	Why
C3	47 pF	loose antenna coupling — keeps tank Q high
L1 CV1	220 μ H + 10-280 pF	the TUNE knob: $\approx$ 614 kHz - 1.8 MHz
C4	100 nF	passes the selected RF, blocks the coil's DC short
R2/R3	1 M / 470 k	bias so light the tank barely notices
R4	2.2 k Ω	Q1's emitter load $\rightarrow$ $\sim$ 1 mA follower current
R8	2.2 k Ω	Q2's collector load — the 90 $\times$ comes from $R8/r_e$

Table 6-1 • The radio-frequency front end. Values chosen so every bias lands mid-range, verified by the 29-check math suite.

6.3 • The detector (D1) — where radio becomes audio

Here is the moment the whole project aims at. C7 couples Q2's amplified RF into D1, a 1N34A germanium diode, and D1 only passes the positive half of every carrier cycle. C8, across the volume control, then fills in the gaps between cycles. What remains across the volume pot is the carrier's *envelope* — and the envelope, chapter 2 told us, is the voice. The component values are a compromise you can hear: C8 and the 10 k Ω pot set a 47 μ s time constant, fast enough to follow a 3 kHz voice crest, slow enough to bridge the 1.3 μ s gaps between 750 kHz carrier peaks. Make C8 much bigger and high voices blur; much smaller and the carrier's own ripple leaks through as a whine. Experiment 10 lets you try both wrong values on purpose.

Germanium is used because its forward drop ($\sim$ 0.2-0.3 V) is gentler than silicon's 0.6 V — with millivolt-to-sub-volt signals, every tenth of a volt of threshold is sensitivity you keep. A Schottky diode (1N5711) makes an excellent modern substitute; a common 1N4148 works but is deaf to weak stations. C7's DC-blocking role matters too: it lets the detector output rest at zero volts with no signal, which is what lets the signal LED below behave as an honest strength meter instead of an always-on lamp.

6.4 • Volume, the signal LED, and the LM386

The volume control is the detector's load itself — a 10 k Ω potentiometer wired as a variable resistor to ground, its wiper picking off a slice of the recovered audio. That slice rides C9 into the LM386, the industry's friendliest audio amplifier chip: eight pins, gain 20 as wired, or 200 when SW2 straps a 10 μ F capacitor across its gain pins. A 220 μ F output capacitor couples it to the 8 Ω speaker, and the 10 Ω + 47 nF Zobel network keeps the chip stable into a real loudspeaker — the LM386's one known vice. At 9 V the chip will happily produce half a watt; the GAIN 20/200 switch exists because a strong local transmitter needs 20 $\times$ while a faint real broadcast station wants all 200 $\times$.

The signal LED is the receiver's party trick. Q3 watches the detector's DC level through R9; when a strong carrier is tuned in, the detector's output climbs past a volt, Q3 turns on, and LED2 glows. Now finding the transmitter is a game with a scoreboard: turn the TUNE knob slowly and watch the light snap on as you cross the station. It is also a genuine signal-strength meter of the kind ham radio operators call an S-meter — the same idea, one LED wide.

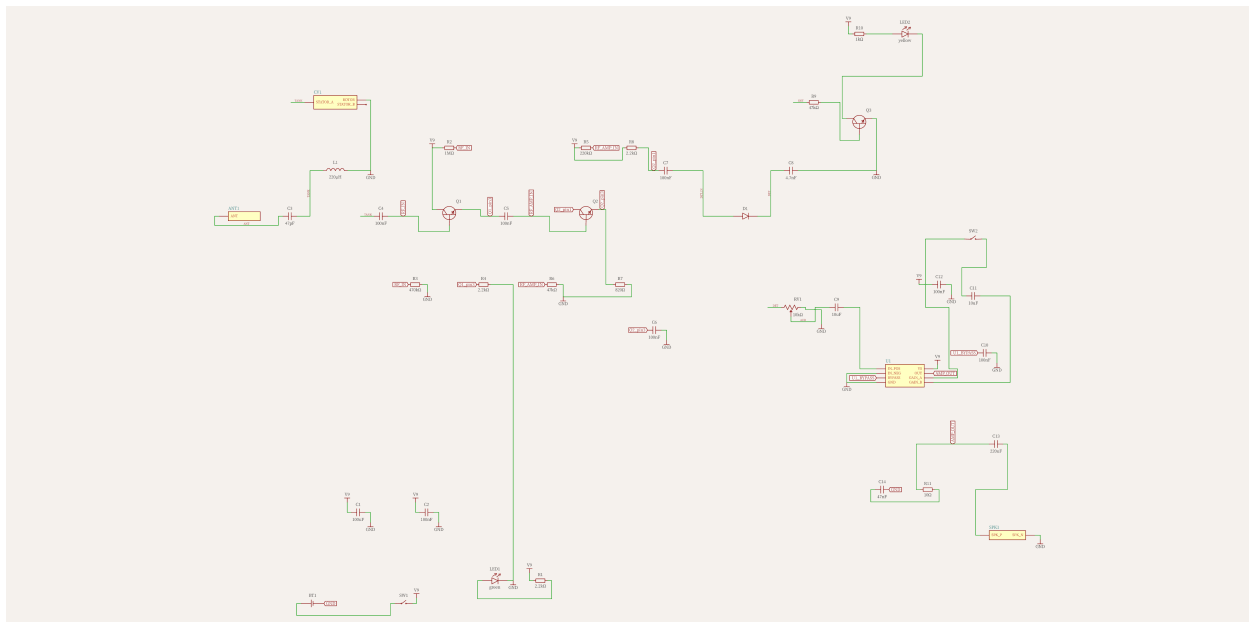


Figure 6-2 • The AM-RX schematic (tscircuit render). Antenna at far left; the detector diode and LM386 occupy the right half.

6.5 • Why not something fancier?

You may know that real radios use the superheterodyne architecture — oscillators and mixers converting everything to a fixed intermediate frequency — and wonder why this design does not. The honest answer: a TRF is the shortest path from antenna to understanding. Its selectivity lives in one tuned circuit you can point at; its detector is one diode you can put a voltmeter on; every signal transformation is visible. The superheterodyne solves problems (mainly: how to get many stations through one sharply tuned amplifier) that a learning radio does not have. Chapter 11 points you at the next steps when you are ready to trade clarity for performance.

7 • Building the pair

7.1 • The shopping list

Every part is through-hole and generic; substitutes are called out where they matter. The two boards share most line items, and a single dual-gang polyvaricon can even be cannibalised to tune both (use one gang per board). Total cost for both boards is typically \$10-20 depending on how stocked your parts drawer already is.

Semiconductors	Qty	Notes / substitutes
BC547B (TO-92)	7	primary transistor; 2N3904 works if inserted rotated (see 7.3)
1N34A germanium diode	1	the detector; 1N5711 Schottky or OA91 fine; 1N4148 deaf-but-works
LM386N (DIP-8)	1	audio power amp, any suffix
LED 5 mm, green	2	power indicators
LED 5 mm, yellow	1	the receiver's signal LED

Table 7-1 • Semiconductors.

Passives & electromechanical	Qty	Notes
Resistors 1/4 W	~25	see value lists below — all 5% or better is fine
Ceramic caps	~20	220p, 47p, 1n, 2.2n, 47n, 100n, 4.7n values used
Electrolytic caps	5	10 μ F $\times$ 2, 100 μ F $\times$ 3 (16 V or 25 V)
Film/MLCC 1 μ F	3	2.54 mm pitch; electrolytic also OK if + faces the mic
Radial chokes	2	100 μ H (TX) and 220 μ H (RX), 5 mm pitch
Polyvaricon 10-280 pF	2	or one dual-gang; any AM tuning cap 10-300 pF works
Potentiometer 10 k Ω lin	1	panel-mount with knob, for volume
Slide/toggle switches	3	power $\times$ 2 + RX gain switch; any SPST/SPDT
Electret mic capsule	1	2-terminal, any size 6-10 mm
Speaker 8 Ω 0.25-0.5 W	1	or headphones (loud!)
9 V battery + clip	2	and a battery snap or holder each
Perfboard 0.1 in pitch	2 pcs	~90 $\times$ 60 and ~100 $\times$ 65 mm, or order the PCBs

Table 7-2 • The rest of the BOM.

Resistor	TX	RX
2.2 k Ω	2	2
10 k Ω	3	1
15 k Ω	1	0
68 k Ω	1	0
4.7 k Ω	1	0
560 Ω	2	0
1 k Ω	1	0
27 k Ω	2	0
1 M Ω	0	1
470 k Ω	0	1
220 k Ω	0	1
47 k Ω	0	2
820 Ω	0	1
10 Ω	0	1

Table 7-3 · Resistor count per board. The shared values are deliberate — one parts drawer serves both.

7.2 · Perfboard or PCB, both work

The PCBs were placed on a strict 2.54 mm grid precisely so the layouts can be copied hole-for-hole onto perfboard: every component position in the renders is a perfboard hole location. If you order the boards instead (the repository exports manufacturer-ready Gerbers for both), you get silkscreen labels, a proper bottom-layer ground pour, and no wiring at all. On perfboard, the ground pour becomes your job — run a bare wire bus along the bottom row of holes and solder every ground to it; at these frequencies a tidy ground bus is not optional polish, it is why the oscillator behaves.

Build order matters on either platform: resistors first (lowest, easiest to turn the board over), then the small capacitors, then diodes and transistors (watch the orientations below), then the big electrolytics, and the mechanical parts — switches, terminal pads, tuning caps — last. Solder the battery snap's wires so the board sits with the schematic's left edge at your left; every debugging step in chapter 8 assumes that orientation.

7.3 · Transistor orientation — the one gotcha

The boards use an in-line TO-92 footprint: three holes in a row, 0.1 in apart. To use it, spread each transistor's legs into a row — the first mechanical skill every kit builder learns. BC547 goes in with its **flat face toward the board edge** (the silkscreen says so); its pins then read Collector-Base-Emitter left to right, matching the board. A 2N3904 has the opposite order (Emitter-Base-Collector), so it goes in **rotated 180°**, flat face inward. The silkscreen prints both instructions on both boards because this single mistake is the classic first-build failure: the circuit looks complete, the LED lights, and nothing transmits — because every

transistor's collector and emitter are swapped.

- **BC547:** flat face toward the board edge, pins C-B-E left to right.
- **2N3904:** flat face the other way (rotated 180°), pins then land C-B-E.
- **1N34A:** the end with the stripe or the visible crystal junction is the cathode — the banded end faces the volume pot, per the board's diode symbol.
- **Electrolytics:** the stripe (minus) side faces ground unless the silkscreen says otherwise; C1 and C13 both have their + toward the +9 V rail.
- **MIC1:** the pin electrically tied to the metal case is MIC_N and faces the ground pad.

7.4 • Wiring the mechanical parts

The battery clips, switches, speaker, and (optionally) the volume pot and tuning capacitors connect through their labelled pads with short stranded wires — keep them under 10 cm. The polyvaricon solders neatly to the board pads directly if you prefer; its two end lugs are the stators, the wide middle lug is the common rotor (ground). On the transmitter, only one stator (A) is used; the spare (B) is a documented upgrade hook — paralleling it adds range at the bottom of the band. The antenna is a 1-2 m wire soldered to the ANT pad; let it trail off the table or tape it up a wall. Nothing about the antenna is critical except its existence: a transmitter with no antenna wire transmits a signal with almost no range at all.

8 • Bring-up: making them talk

Bring-up is a sequence of small verifications, each with a pass condition you can see or hear. Do not skip ahead: each step confirms the pieces the next step depends on. The only instrument you might want is a pocket AM radio or a phone with an AM app — and even that is optional, because the receiver you just built is the test equipment for the transmitter, and vice versa.

#	Action	What success looks like
1	TX board: battery in, switch on	green PWR LED lights, ~6.5 mA draw
2	RX board: battery in, switch on	green PWR LED lights, faint hiss with volume up
3	RX: GAIN switch open (20×), volume mid	hiss level changes with volume — audio chain alive
4	RX: TUNE slowly across the whole band	you should catch real broadcast stations (evening is best)
5	TX: set TUNE mid-range	(all quiet so far — keep going)
6	RX: sweep TUNE slowly around the TX range	static → voice → static; yellow SIG LED glows at the peak
7	Speak into the mic from ~10 cm	your voice, recognisably, from the speaker
8	Move the boards apart in steps	range fades over a few rooms — that's the physics of chapter 2

Table 8-1 • The bring-up sequence. Steps 1-5 validate each board alone; 6-8 validate the link.

8.1 • Step 4 deserves a paragraph of its own

Tuning across the band and hearing a real broadcast station — VOV1 on 630 or 675 kHz in most of Vietnam, or whatever your region offers after dark — is the best single test that the receiver's front end works, and it needs no transmitter at all. Medium-wave reception improves dramatically after sunset, when the ionosphere's D-layer stops absorbing the skywave; if your first attempt is at noon and hears only locals, try again in the evening with the same settings. Lengthening the antenna wire to 5-10 m (out the window, along a fence) transforms weak-station reception and is the single cheapest upgrade the receiver will ever get.

8.2 • The tuning game

Once the link works, play the game the boards were designed around: park the receiver on a quiet spot between stations, then tune the *transmitter* until its carrier appears — the SIG LED becomes your scoreboard. Now walk the transmitter across the band and follow it with the receiver. You are watching a station and a listener chase each other across the dial, which is the relationship every radio in your house has been having silently all along. When the two frequencies coincide, you will also hear a slow beat note as they *almost* coincide — the same beat phenomenon that makes two slightly-out-of-tune guitar strings warble, now happening at radio scale.

**carrier → beat →
lock**

what you hear as the two TUNE knobs
converge — the beat note is the audible proof
that two oscillators are involved

9 • Twelve experiments

Each of these changes exactly one thing and predicts what you will observe. That is the experimental method, worn casually. Do them in any order; several want a helper to talk continuously into the mic while you listen.

#	Experiment	One-line prediction
1	Distance ladder: 1 m, 3 m, 10 m, next room	loudness falls fast — near-field's $1/d^2$ in action
2	Shout vs whisper into the mic	shout distorts — you crossed $m = 1$ into overmodulation
3	Remove the antenna wire	range collapses to centimetres
4	Double the antenna (4 m)	range grows, but not double — diminishing returns
5	Tune TX onto a real broadcast station's frequency	your voice and theirs mix — interference, live
6	Touch the RX antenna with your hand	station shifts — your body adds capacitance to the tank
7	Swap the 1N34A for a 1N4148	weak stations fade — silicon's higher threshold
8	Close the RX GAIN switch (200×)	weak stations appear; the TX now saturates — gain is a choice
9	Solder 4.7 kΩ across the RX tank	stations smear together — Q collapsed, selectivity gone
10	Change C8 to 100 nF (detector cap)	voices blur — the envelope filter got too slow
11	Parallel the TX varcap's spare gang (A+B)	range extends downward — more C, lower f_0
12	Listen at night, antenna out the window	distant stations bloom — the ionosphere woke up

Table 9-1 · The experiment menu. Each row is a hypothesis you can test in minutes.

9.1 · The three that teach the most

If you only do three, do 2, 9 and 10. Experiment 2 makes the invisible visible: overmodulation distortion is the sound of the carrier momentarily vanishing, and once you have heard it you will recognise the same artifact in every badly-run pirate station and cheap intercom. Experiment 9 is selectivity made visceral — with the tank loaded, the knob stops choosing stations and the band becomes soup, which is exactly what the buffer's 1 MΩ bias pair exists to prevent. Experiment 10 lets you feel the detector's compromise between carrier ripple and audio fidelity — the 47 μs time constant chosen in chapter 6 stops being a number and becomes a sound.

Write down what you observe, especially where it differs from the prediction. A prediction that fails is not a broken project; it is a live circuit telling you which simplification in your mental model was too bold. The engineers who built radio spent fifty years collecting exactly such surprises — you get to collect a few of the best ones in an afternoon.

10 • Troubleshooting

Fault-finding a radio you built yourself is the fastest electronics education there is, provided you bisect instead of guess: test the chain in order, and let each test cut the possible faults in half. The tables below map symptoms to first suspects, board by board. A multimeter helps but is not required — the signal LED, the speaker, and a pocket radio make a complete test bench for this project.

Symptom (TX)	First suspects, in order
PWR LED dark	battery, switch wiring, LED polarity
No carrier heard on any radio	transistor orientation (7.3!); L1's pad 2 must go to VMD, not ground
Carrier but no voice	mic polarity (case pin to ground); C3/C5 cold joint; speak at 10 cm
Voice badly distorted	overmodulation (back off / whisper); electret shorted against its case
Carrier drifts or warbles	C6 accidentally 100 nF (audio into the bias); battery below 7 V
Carrier only at one end of TUNE	varcap lugs miswired (rotor = middle/wide lug to ground)

Table 10-1 • Transmitter symptoms.

Symptom (RX)	First suspects, in order
PWR LED dark	battery, switch wiring
No hiss at any volume	LM386 wiring; C13 polarity; speaker pads bridged
Hiss but no stations, ever	tank wiring (L1/CV1), antenna wire missing, D1 backwards
Hears locals but not the TX	TX on a frequency outside 614-836 kHz — retune TX mid-band
SIG LED always on	Q3 collector/emitter swapped; detector output shorted to V9
Stations all smeared together	experiment 9's resistor left in; heavy antenna (C3 too big)
Buzz at 50/60 Hz everywhere	battery gone flat; ground bus skipped on perfboard

Table 10-2 • Receiver symptoms.

10.1 • The two-minute sanity loop

When lost, return to the chain diagrams: figure 4-1 for the transmitter, figure 6-1 for the receiver. Test each stage's *output promise* in order — power rail present? oscillator running (a nearby AM radio hisses/squeals at the carrier even with no voice)? detector DC jumping when a station arrives? audio hiss present? — and the first broken promise is your fault region. This is the same discipline the verification suite applies to the design (chapter 5), now applied with your ears to the physical board. Design, simulation, and debugging are one habit applied at three altitudes.

11 • The rules, the safety, and the road ahead

11.1 • Transmitting legally

Low-power unlicensed transmitters are regulated, but generously, because their range is intrinsically small. In the United States, FCC Part 15.219 permits an unlicensed AM transmitter with up to 100 mW of DC input power to its final stage and an antenna plus ground lead no longer than 3 metres. This transmitter draws about 5 mW into its final — twenty times under the ceiling — and a 2 metre wire antenna keeps you inside the antenna rule. Many other countries have comparable provisions for field strengths this small; a few require notification, and a handful are stricter. Two habits keep you polite everywhere: keep the antenna short, and choose a quiet frequency between broadcast stations rather than on top of one. Receiving, of course, is legal everywhere and always was.

11.2 • Safety, briefly

A 9 V battery cannot hurt you, and nothing in these boards stores meaningful energy — the highest voltage anywhere is the battery itself. The real risks are the timeless bench ones: soldering iron burns, rosin fumes (ventilate), and leads clipped short flying into eyes (glasses). If you later upgrade the transmitter with amplifier modules or higher rails, re-read this paragraph — the friendly numbers here belong to a 9 V battery, not to whatever you build next.

11.3 • Where this road goes next

You now own, in your head, the skeleton of every receiver ever made: antenna, tuning, amplification, detection, audio. Three doors lead onward. The **regenerative receiver** adds one transistor of positive feedback to the TRF and achieves startling sensitivity — Armstrong's 1912 idea, still a favourite build. The **superheterodyne** is the architecture of virtually every real radio: a local oscillator and mixer convert every station to one fixed frequency where sharp filters live; build one and you will finally understand the numbers printed on every radio dial. And **software-defined radio** replaces the tuned circuits with mathematics — a \$30 USB stick lets you watch the entire spectrum you have been playing in, as a waterfall, in real time. Whichever door you take, you will keep using the ideas this pair of boards made physical: resonance selects, nonlinearity creates, and every signal you will ever chase is someone's deliberate ride on a carrier.

**You built both
ends**

the station and the listener

**You proved it
first**

29 math checks · 7 simulation proofs

**You tuned the
ether**

the knob really does slide the bell